

ECE 332 Fall 2025 — Exam 1 Answer Key

[CSp.1: SECTION] = page 1 of cheat sheet [CSp.2: SECTION] = page 2

Two corrected errors from prior keys:

(1) **Q1(c):** Exam writes “ $\omega = 2 \text{ MHz}$ ” but MHz is a *frequency* unit. Must convert: $\omega = 2\pi f = 2\pi \times 2 \times 10^6 = 4\pi \times 10^6 \text{ rad/s}$.

(2) **Q6:** Exam gives $f = \frac{2}{\pi} \text{ GHz}$ (not $\frac{2}{5} \text{ GHz}$). $\omega = 2\pi f = 2\pi \cdot \frac{2}{\pi} \times 10^9 = 4 \times 10^9 \text{ rad/s}$.

Question	Topic	Points	Page
Q1	Triangular loop EMF	20	2
Q2	Moving wire (motional EMF)	12	3
Q3	Conduction & displacement current	12	4
Q4	Charge relaxation	10	5
Q5	Lossless plane wave	30	6
Q6	Lossy wave in distilled water	16	7
Total		100	

Question 1 — Triangular Loop

20 pts

Equilateral loop in y - z plane, N turns, side a . $\mathbf{B}(t) = B_0(-2\hat{x} + 3\hat{y})\sin(\omega t)$.

Cheat Sheet Snippets (all p.1)

EMF & FARADAY'S LAW

$$V_{\text{emf}} = -N \frac{d\Phi}{dt} \quad \Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$$

DOT PRODUCTS

$$\hat{x} \cdot \hat{x} = 1, \quad \hat{y} \cdot \hat{x} = 0$$

 $\partial B / \partial t$ TABLE

$$\frac{d}{dt} \sin(\omega t) = \omega \cos(\omega t)$$

UNIT CIRCLE

$$\cos(\pi/4) = \frac{\sqrt{2}}{2}$$

SURFACE ELEMENT $d\mathbf{s}$

$$y\text{-}z \text{ plane} \Rightarrow d\mathbf{s} = \hat{x} da$$

ROTATING LOOP — areas

$$\text{Equil. triangle (side } a\text{): } A = \frac{\sqrt{3}}{4} a^2$$

CIRCUIT & WAVE strip

$$\omega = 2\pi f$$

(a) Magnetic flux through one turn.

(6 pts)

$$\begin{aligned} \Phi &= \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S B_0(-2\hat{x} + 3\hat{y})\sin(\omega t) \cdot \hat{x} da \\ &= A \left(-2 \underbrace{(\hat{x} \cdot \hat{x})}_1 + 3 \underbrace{(\hat{y} \cdot \hat{x})}_0 \right) B_0 \sin(\omega t) = -2 \cdot \frac{\sqrt{3}}{4} a^2 \cdot B_0 \sin(\omega t) \end{aligned}$$

$$\Phi = -\frac{\sqrt{3}}{2} B_0 a^2 \sin(\omega t) \text{ V}\cdot\text{s}$$

(b) EMF induced in the loop.

(6 pts)

$$V_{\text{emf}} = -N \frac{d}{dt} \left(-\frac{\sqrt{3}}{2} a^2 B_0 \sin(\omega t) \right) = \frac{\sqrt{3}}{2} N a^2 B_0 \cdot \omega \cos(\omega t)$$

$$V_{\text{emf}}(t) = \frac{\sqrt{3}}{2} a^2 B_0 N \omega \cos(\omega t) \text{ V}$$

(c) Current through $1 \text{ k}\Omega$ at given values.

(8 pts)

Exam says “ $\omega = 2 \text{ MHz}$ ” $\Rightarrow$ convert first [CS p.1: CIRCUIT & WAVE]: $\omega = 2\pi f = 2\pi(2 \times 10^6) = 4\pi \times 10^6 \text{ rad/s}$

$N = 10$, $a = 0.08 \text{ m}$, $B_0 = 0.02 \text{ T}$, $\omega = 4\pi \times 10^6 \text{ rad/s}$, $\omega t = \pi/4$, $R = 1000 \Omega$.

$$|V_{\text{emf}}| = \frac{\sqrt{3}}{2} (0.08)^2 (10) (0.02) (4\pi \times 10^6) \underbrace{\cos\left(\frac{\pi}{4}\right)}_{\sqrt{2}/2} = 9.85 \times 10^3 \text{ V}$$

$$I = \frac{V_{\text{emf}}}{R} = \frac{9.85 \text{ kV}}{1000 \Omega} \quad \boxed{I \approx 9.85 \text{ A}}$$

Question 2 — Moving Wire

12 pts

$a = 0.15 \text{ m}$ along $\hat{y}$, $\mathbf{u} = 0.12\hat{x} \text{ m/s}$, $\mathbf{B} = (0.1\hat{x} - 0.05\hat{z}) \text{ T}$.

Cheat Sheet Snippets (p.1)

EMF & FARADAY'S LAW (Motional) CROSS PRODUCT cycle

$$V_{\text{emf}} = \oint (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$

$d\mathbf{l} = \hat{y} dy$ along a $\hat{y}$ -wire

$$\hat{x} \times \hat{z} = -\hat{y} \quad (\text{reversed cycle})$$

$$\hat{x} \times \hat{x} = 0 \quad (\text{same vectors})$$

$$\mathbf{u} \times \mathbf{B} = 0.12\hat{x} \times (0.1\hat{x} - 0.05\hat{z}) = \underbrace{0.12 \times 0.1}_{=0}(\hat{x} \times \hat{x}) + \underbrace{0.12 \times (-0.05)}_{=-0.006}(\hat{x} \times \hat{z})$$

$$= 0 + (-0.006)(-\hat{y}) = 6 \times 10^{-3} \hat{y} \text{ V/m}$$

$$V_{\text{emf}} = \int_0^{0.15} (6 \times 10^{-3} \hat{y}) \cdot (\hat{y} dy) = 6 \times 10^{-3} (0.15)$$

$$V_{\text{emf}} = 900 \mu\text{V} = 0.9 \text{ mV}$$

Question 3 — Conduction & Displacement Current

12 pts

$$I_c = 2 \sin(\omega t) \mu\text{A}, \quad \sigma = 10^{-4} \text{ S/m}, \quad \varepsilon_r = 26\pi.$$

Cheat Sheet Snippets (p.1)

DISPLACEMENT & CONDUCTION CURRENT UNIT CIRCLE

$$E = V(t)/d \quad I_c = \frac{\sigma A}{d} V(t) \quad J_c = \sigma E \quad \cos(\pi/3) = \frac{1}{2}$$

$$I_d = C \frac{dV}{dt} \quad C = \frac{\varepsilon_r \varepsilon_0 A}{d} \quad J_d = \varepsilon \frac{\partial E}{\partial t}$$

(a) Electric field E .

(6 pts)

$$E = \frac{V(t)}{d}, \quad I_c = \frac{\sigma A}{d} V(t) \Rightarrow V(t) = \frac{I_c d}{\sigma A}$$

$$E = \frac{1}{d} \cdot \frac{I_c d}{\sigma A} = \frac{I_c}{\sigma A} = \frac{2 \times 10^{-6} \sin(\omega t)}{10^{-4} \cdot A} \quad \boxed{E = \frac{0.02}{A} \sin(\omega t) \text{ V/m}}$$

(b) Displacement current I_d at $\omega = 10^9 \text{ rad/s}$, $\omega t = \pi/3$.

(6 pts)

From (a): $E = V(t)/d \Rightarrow V(t) = \frac{d \cdot 0.02}{A} \sin(\omega t)$.

$$I_d = \frac{\varepsilon_r \varepsilon_0 A}{d} \cdot \frac{dV}{dt} = \frac{\varepsilon_r \varepsilon_0 A}{d} \cdot \frac{d}{dt} \left(\frac{d \cdot 0.02}{A} \sin(\omega t) \right) = \varepsilon_r \varepsilon_0 \cdot 0.02 \cdot \omega \cos(\omega t)$$

At $\omega t = \pi/3$, $\cos(\pi/3) = \frac{1}{2}$:

$$I_d = (26\pi)(8.85 \times 10^{-12})(0.02)(10^9)\left(\frac{1}{2}\right) \quad \boxed{I_d \approx 7.23 \text{ mA}}$$

Question 4 — Charge Relaxation**10 pts**

$$\sigma = 10^{-4} \text{ S/m}, \quad \varepsilon_r = 26\pi.$$

Cheat Sheet Snippet (p.1 — CHARGE RELAXATION)

$$\tau_r = \frac{\varepsilon_r \varepsilon_0}{\sigma} \qquad \rho_v(t) = \rho_{v0} e^{-t/\tau_r}$$

(a) Relaxation time constant.*(5 pts)*

$$\tau_r = \frac{\varepsilon_r \varepsilon_0}{\sigma} = \frac{(26\pi)(8.85 \times 10^{-12})}{10^{-4}} \quad \boxed{\tau_r \approx 7.23 \mu\text{s}}$$

(b) Time to decay to e^{-3} of initial value.*(5 pts)*Set $\rho_{v0} e^{-3} = \rho_{v0} e^{-t/\tau_r}$, take $\ln$ both sides:

$$\ln e^{-3} = \frac{-t}{\tau_r} \Rightarrow t = -\tau_r \ln(e^{-3})$$

$$\boxed{t = 3 \times 7.23 \mu\text{s} \approx 21.69 \mu\text{s}}$$

Question 5 — Lossless EM Wave

30 pts

$$\mathbf{E}(z, t) = 4 \cos\left((6\pi \times 10^8)t - kz - \frac{\pi}{6}\right) \hat{y} \text{ V/m. } \epsilon_r = 6, \mu_r = 1.$$

Cheat Sheet Snippets

WAVE PARAMS (p.2 bottom strip)

$$k = \omega \sqrt{\mu_r \mu_0 \epsilon_r \epsilon_0}, \quad \lambda = \frac{2\pi}{k}$$

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{377}{\sqrt{\epsilon_r}}$$

PHASOR METHOD (p.1)

$$\mathbf{E} = \text{Re}\{\tilde{\mathbf{E}} e^{j\omega t}\}; \text{ strip } e^{j\omega t}$$

PHASOR $\leftrightarrow$ TIME (p.2)

$$\text{Re}\{A e^{j(\omega t - \beta z + \phi)}\} = A \cos(\omega t - \beta z + \phi)$$

DIRECTION OF PROP. (p.2)

$$-kz \text{ in arg} \Rightarrow \hat{k} = +\hat{z}$$

$$\hat{k} \times \hat{E} \text{ TABLE (p.2): } +\hat{z}, \hat{y} \Rightarrow -\hat{x}$$

LOSSLESS E & H (p.2)

$$\tilde{\mathbf{H}} = \frac{1}{\eta} (\hat{k} \times \tilde{\mathbf{E}})$$

POYNTING lossless (p.2)

$$\mathbf{S}_{\text{av}} = \frac{|E_0|^2}{2\eta} \hat{k}$$

(a) ω, k, λ, η .

(10 pts)

$$\boxed{\omega = 6\pi \times 10^8 \text{ rad/s}} \quad k = (6\pi \times 10^8) \sqrt{(4\pi \times 10^{-7})(6)(8.85 \times 10^{-12})} \quad \boxed{k \approx 15.40 \text{ rad/m}}$$

$$\lambda = \frac{2\pi}{15.40} \quad \boxed{\lambda \approx 0.408 \text{ m}} \quad \eta = \frac{377}{\sqrt{6}} \quad \boxed{\eta \approx 153.91 \Omega}$$

(b) $\tilde{\mathbf{E}}(z)$.

(5 pts)

Let $\phi = \pi/6$. Strip $e^{j\omega t}$: $\tilde{\mathbf{E}} = 4 e^{j(-kz - \phi)} \hat{y}$.

$$\boxed{\tilde{\mathbf{E}}(z) = 4 e^{-j(15.40z + \pi/6)} \hat{y} \text{ V/m}}$$

(c) $\tilde{\mathbf{H}}(z)$.

(5 pts)

$$\tilde{\mathbf{H}} = \frac{1}{\eta} (\hat{z} \times \tilde{\mathbf{E}}) = \frac{1}{\eta} (\hat{z} \times 4 e^{-j(15.40z + \pi/6)} \hat{y}) = -\frac{4}{\eta} e^{-j(15.40z + \pi/6)} \hat{x} = -\frac{4}{153.91} e^{-j(15.40z + \pi/6)} \hat{x}$$

$$\boxed{\tilde{\mathbf{H}}(z) \approx -25.99 \times 10^{-3} e^{-j(15.40z + \pi/6)} \hat{x} \text{ A/m}}$$

(d) $\mathbf{H}(z, t)$.

(5 pts)

$$\begin{aligned} \mathbf{H}(z, t) &= \text{Re}\left\{-25.99 \times 10^{-3} e^{-j(15.40z + \pi/6)} \hat{x} \cdot e^{j\omega t}\right\} \\ &= \text{Re}\left\{-25.99 \times 10^{-3} e^{j(\omega t - 15.40z - \pi/6)} \hat{x}\right\} \end{aligned}$$

$$\boxed{\mathbf{H}(z, t) = -25.99 \times 10^{-3} \cos\left(6\pi \times 10^8 t - 15.40z - \frac{\pi}{6}\right) \hat{x} \text{ A/m}}$$

(e) Average power density.

(5 pts)

$$\text{Lossless} \Rightarrow \mathbf{S}_{\text{av}} = \frac{|E_0|^2}{2\eta} \hat{k}:$$

$$\mathbf{S}_{\text{av}} = \frac{4^2}{2(153.91)} \hat{z} \quad \boxed{\mathbf{S}_{\text{av}} \approx 51.98 \times 10^{-3} \hat{z} \text{ W/m}^2}$$

Question 6 — EM Wave in Distilled Water

16 pts

$$f = \frac{2}{\pi} \text{ GHz}, \quad \sigma = 10^{-4} \text{ S/m}, \quad \epsilon_r = 26\pi, \quad \mu_r = 1.$$

Cheat Sheet Snippets (p.2)

WAVE PARAMS strip

$$\omega = 2\pi f$$

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{377}{\sqrt{\epsilon_r}}$$

SKIN DEPTH — LOW-LOSS MEDIUM

$$\delta_s = \frac{2}{\sigma\eta} \quad (\text{from } \alpha = \sigma\eta/2, \delta_s = 1/\alpha)$$

LOSSY MEDIA — CLASSIFICATION

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega\epsilon_r\epsilon_0}$$

 $< 0.01 \Rightarrow \text{low-loss dielectric}$

$$\omega = 2\pi f = 2\pi \cdot \frac{2}{\pi} \times 10^9 = 4 \times 10^9 \text{ rad/s}$$

(a) Find ϵ''/ϵ' .

(6 pts)

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega\epsilon_r\epsilon_0} = \frac{10^{-4}}{(4 \times 10^9)(26\pi)(8.85 \times 10^{-12})}$$

$$\frac{\epsilon''}{\epsilon'} \approx 34.58 \times 10^{-6}$$

(b) What type of medium?

(5 pts)

Since $34.58 \times 10^{-6} \ll 0.01$, at this frequency distilled water is a **low-loss dielectric**.

(c) Skin depth.

(5 pts)

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{377}{\sqrt{26\pi}} = \frac{377}{9.038} \approx 41.71 \, \Omega$$

$$\delta_s = \frac{2}{\sigma\eta} = \frac{2}{(10^{-4})(41.71)} \quad \boxed{\delta_s \approx 479.5 \text{ m}}$$

Score (if perfect): Q1: 20 — Q2: 12 — Q3: 12 — Q4: 10 — Q5: 30 — Q6: 16 = **100/100**